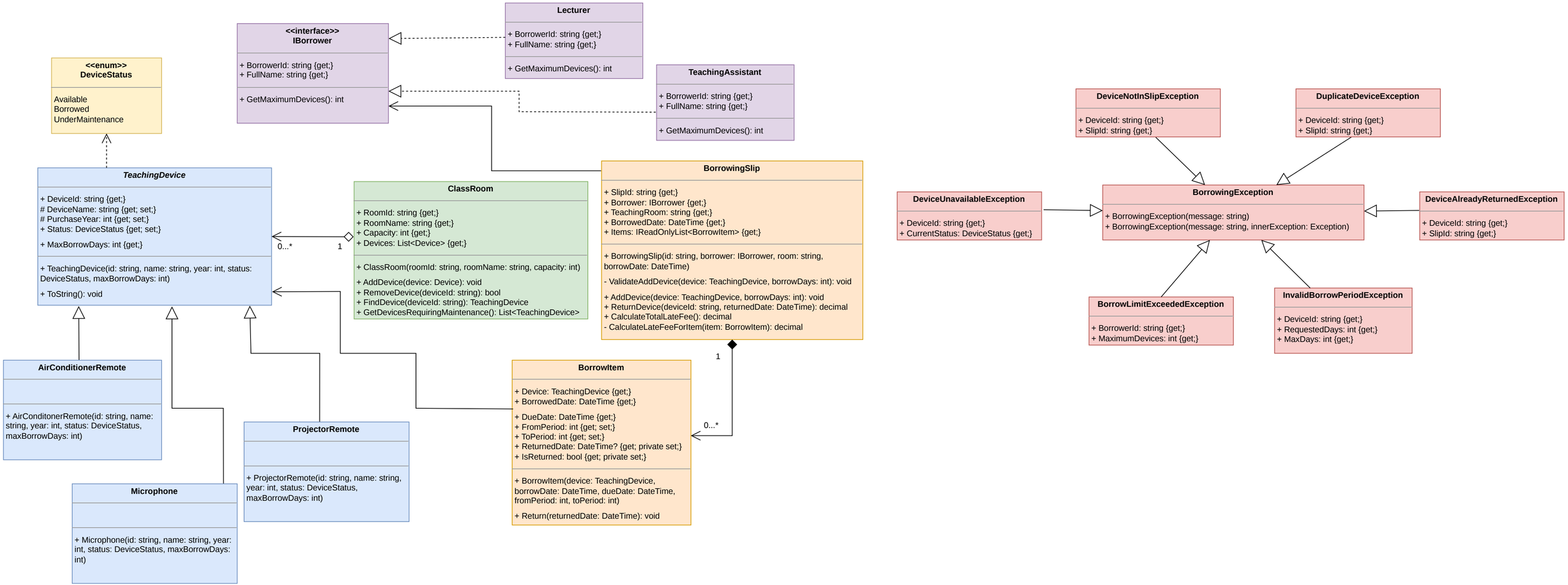
<<interface>>
IBorrower
+ BorrowerId: string {get;}
+ FullName: string {get;}
+ GetMaximumDevices(): int
Lecturer
+ BorrowerId: string {get;}
+ FullName: string {get;}
+ GetMaximumDevices(): int
TeachingAssistant
+ BorrowerId: string {get;}
+ FullName: string {get;}
+ GetMaximumDevices(): int
<<enum>>
DeviceStatus
Available
Borrowed
UnderMaintenance
TeachingDevice
+ DeviceId: string {get;}
# DeviceName: string {get; set;}
# PurchaseYear: int {get; set;}
+ Status: DeviceStatus {get; set;}
+ MaxBorrowDays: int {get;}
+ TeachingDevice(id: string, name: string, year: int, status: DeviceStatus, maxBorrowDays: int)
+ ToString(): void
ClassRoom
+ RoomId: string {get;}
+ RoomName: string {get;}
+ Capacity: int {get;}
+ Devices: List<Device> {get;}
+ ClassRoom(roomId: string, roomName: string, capacity: int)
+ AddDevice(device: Device): void
+ RemoveDevice(deviceId: string): bool
+ FindDevice(deviceId: string): TeachingDevice
+ GetDevicesRequiringMaintenance(): List<TeachingDevice>
0...*
1
BorrowingSlip
+ SlipId: string {get;}
+ Borrower: IBorrower {get;}
+ TeachingRoom: string {get;}
+ BorrowedDate: DateTime {get;}
+ Items: IReadOnlyList<BorrowItem> {get;}
+ BorrowingSlip(id: string, borrower: IBorrower, room: string, borrowDate: DateTime)
- ValidateAddDevice(device: TeachingDevice, borrowDays: int): void
+ AddDevice(device: TeachingDevice, borrowDays: int): void
+ ReturnDevice(deviceId: string, returnedDate: DateTime): decimal
+ CalculateTotalLateFee(): decimal
- CalculateLateFeeForItem(item: BorrowItem): decimal
1
0...*
BorrowItem
+ Device: TeachingDevice {get;}
+ BorrowedDate: DateTime {get;}
+ DueDate: DateTime {get;}
+ FromPeriod: int {get; set;}
+ ToPeriod: int {get; set;}
+ ReturnedDate: DateTime? {get; private set;}
+ IsReturned: bool {get; private set;}
+ BorrowItem(device: TeachingDevice, borrowDate: DateTime, dueDate: DateTime, fromPeriod: int, toPeriod: int)
+ Return(returnedDate: DateTime): void
AirConditionerRemote
+ AirConditonerRemote(id: string, name: string, year: int, status: DeviceStatus, maxBorrowDays: int)
ProjectorRemote
+ ProjectorRemote(id: string, name: string, year: int, status: DeviceStatus, maxBorrowDays: int)
Microphone
+ Microphone(id: string, name: string, year: int, status: DeviceStatus, maxBorrowDays: int)
DeviceNotInSlipException
+ DeviceId: string {get;}
+ SlipId: string {get;}
DuplicateDeviceException
+ DeviceId: string {get;}
+ SlipId: string {get;}
DeviceUnavailableException
+ DeviceId: string {get;}
+ CurrentStatus: DeviceStatus {get;}
BorrowingException
+ BorrowingException(message: string)
+ BorrowingException(message: string, innerException: Exception)
DeviceAlreadyReturnedException
+ DeviceId: string {get;}
+ SlipId: string {get;}
BorrowLimitExceededException
+ BorrowerId: string {get;}
+ MaximumDevices: int {get;}
InvalidBorrowPeriodException
+ DeviceId: string {get;}
+ RequestedDays: int {get;}
+ MaxDays: int {get;}